

From noise to knowledge: how randomness generates novel phenomena and reveals information

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Abstract

Noise, as the term itself suggests, is most often seen a nuisance to ecological insight, a inconvenient reality that must be acknowledged, a haystack that must be stripped away to reveal the processes of interest underneath. Yet despite this well-earned reputation, noise is often interesting in its own right: noise can induce novel phenomena that could not be understood from some underlying deterministic model alone. Nor is all noise the same, and close examination of differences in frequency, color or magnitude can reveal insights that would otherwise be inaccessible. Yet with each aspect of stochasticity leading to some new or unexpected behavior, the time is right to move beyond the familiar refrain of “everything is important” (Bjørnstad & Grenfell 2001). Stochastic phenomena can suggest new ways of inferring process from pattern, and thus spark more dialog between theory and empirical perspectives that best advances the field as a whole. I highlight a few compelling examples, while observing that the study of stochastic phenomena are only beginning to make this translation into empirical inference. There are rich opportunities at this interface in the years ahead.

Keywords: Coloured noise, demographic noise, environmental noise, quasi-cycles, stochasticity, tipping points

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1. Bibliography styles

Here are two sample references: [Feynman and Vernon Jr. \(1963\)](#) [Dirac \(1953\)](#).

With this template using `elsevier` class, `natbib` will be used. Three bibliographic style files (*.bst) are provided and their use controlled by `cite-style` option:

- `citestyle: number` (default) will use `elsarticle-num.bst` - can be used for the numbered scheme
- `citestyle: numbername` will use `elsarticle-num-names.bst` - can be used for numbered with new options of `natbib.sty`
- `citestyle: authoryear` will use `elsarticle-harv.bst` — can be used for author year scheme

This `citestyle` will insert the right .bst and set the correct `classoption` for `elsarticle` document class.

Using `natbiboptions` variable in YAML header, you can set more options for `natbib` itself. Example

```
natbiboptions: longnamesfirst,angle,semicolon
```

1.1. Using CSL

If `cite-method` is set to `citeproc` in `elsevier_article()`, then `pandoc` is used for citations instead of `natbib`. In this case, the `cs1` option is used to format the references. By default, this template will provide an appropriate style, but alternative `cs1` files are available from <https://www.zotero.org/styles?q=elsevier>. These can be downloaded and stored locally, or the url can be used as in the example header.

2. Equations

Here is an equation:

$$f_X(x) = \left(\frac{\alpha}{\beta}\right) \left(\frac{x}{\beta}\right)^{\alpha-1} e^{-\left(\frac{x}{\beta}\right)^\alpha}; \alpha, \beta, x > 0.$$

Inline equations work as well: $\sum_{i=2}^{\infty} \{\alpha_i^\beta\}$

3. Figures and tables

Figure [1](#) is generated using an R chunk.

4. Tables coming from R

Tables can also be generated using R chunks, as shown in Table [1](#) example.

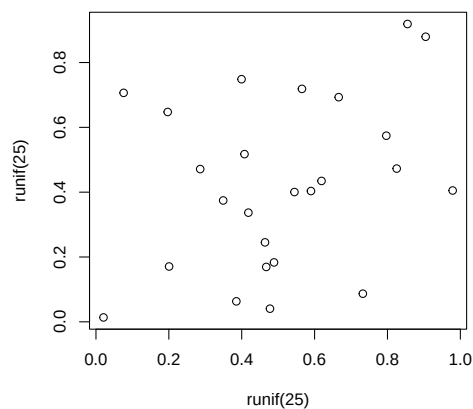


Figure 1: A meaningless scatterplot

```
knitr::kable(head(mtcars)[,1:4])
```

Table 1: Caption centered above table

	mpg	cyl	disp	hp
Mazda RX4	21.0	6	160	110
Mazda RX4 Wag	21.0	6	160	110
Datsun 710	22.8	4	108	93
Hornet 4 Drive	21.4	6	258	110
Hornet Sportabout	18.7	8	360	175
Valiant	18.1	6	225	105

References

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- Feynman, R.P., Vernon Jr., F.L., 1963. The theory of a general quantum system interacting with a linear dissipative system. *Annals of Physics* 24, 118–173. doi:[10.1016/0003-4916\(63\)90068-X](https://doi.org/10.1016/0003-4916(63)90068-X).